Birmingham River Champion's
Water Chemistry Guidance

Water chemistry monitoring is an important way of monitoring river health. Citizen science data on water chemistry play a vital role in detecting pollution by tracking changes over time, helping protect aquatic life. In Birmingham River Champions, we provide equipment to measure different water quality elements. This guidance outlines why different elements are measured and how monitoring should be carried out.

Your water chemistry tests will include:

- A Turbidity Tube for testing how clear or cloudy the water is
- A digital sensor to record water tempurature and electrical conductivity
- nitrate and phospohate testing kits

We suggest all water quality samples are taken in slightly deeper, slower waters (making sure you are not entering above welly height); this could be immediately upstream or downstream of the riffle where you collect Urban Riverfly samples. You can collect your water sample using a container like a jug, glass jar or even the turbidity tube, from which all of the necessary testing can be performed. Ensure that the container is swilled out with river water twice before testing. Water should be collected just below the surface (10-30 cm deep) - avoid sampling only surface water and disturbing the underlying sediment where possible.

Electrical conductivity and temperature

Electrical conductivity shows how easily electricity can pass through water, which depends on the amount of dissolved salts or minerals in the water. Low concentrations of salts naturally occur in rivers, but higher levels can occur due to pollution sources like road salt, wastewater, or industrial runoff. Electrical conductivity is most effective at detecting pollution sources during lower flow conditions when they are less diluted.



Water temperature is crucial for freshwater life, regulating behaviours (e.g., migratory patterns), biological processes (e.g., metabolic rates), and controlling dissolved oxygen levels. Temperature changes can occur below industrial or sewage discharges.

Electrical conductivity and water temperature can both be measured via the **HM Digital sensor**. For this, simply turn the sensor on. Electrical conductivity should be the default measurement ('µs'), and then click the 'Temp' button to change to temperature. You can keep clicking this button to alternate between the different measures.

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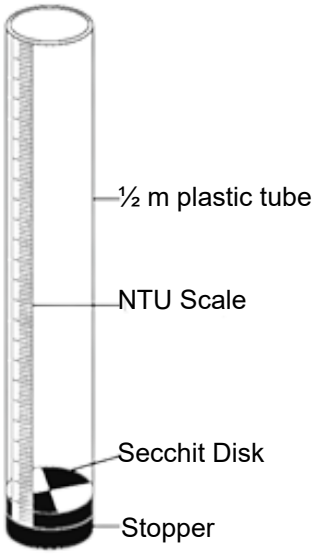
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Turbidity

Turbidity measures how clear or cloudy the water is. Cloudy or murky water (or water with high turbidity) has more particles in the water, e.g. microscopic algae or very fine sediments.

Turbidity can be caused by erosion, runoff (particularly during floods) or unnatural contaminants from industry, roads, sewage or agriculture. High turbidity levels *can* indicate poor water quality as it stops light passing through the water to plants and species that depend on light to survive (although, for some rivers, it can be normal).

For BRC, turbidity is measured using Freshwater Watch kits. A ½ metre plastic tube with a Secchi disc (a disc with black and white triangles that are easy to see) at the bottom is used, with a 'Nephelometric Turbidity Units' (NTU) scale on the side. This is your turbidity tube.



Turbidity measurements are then based on the depth of water in the plastic tube after which you can no longer see the separation between the disc's black and white triangles, measured in NTU. You can either add water gradually into the tube until the triangles can't be separated, or fill the tube in its entirety and empty gradually until the triangles can be separated. Remember to use the same technique each time!

Nitrate and Phosphate

Nitrates and phosphates are nutrients essential for plant growth, but too much of these nutrients can cause an increased growth of algae, leading to harmful algal blooms that can reduce dissolved oxygen in the water needed for fish and insects to survive ('eutrophication').

High nitrate and/or phosphate levels can stem from agricultural, sewage or industrial pollution. Nitrate and phosphate levels are being assessed via sampling kits provided by Freshwater Watch, and your data will contribute to this global initiative alongside Birmingham River Champions.

Testing involved using nutrient-testing tubes which contain a chemical reagent that changes colour when it comes into contact with either nitrate or phosphate. The intensity of the colour depends on the concentration of dissolved nitrate or phosphate in the sample water.

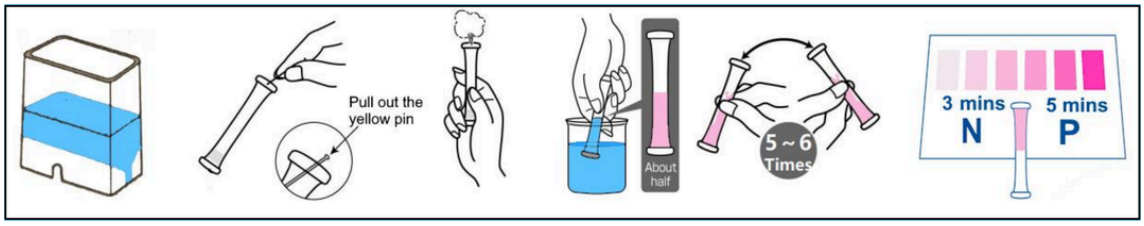
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Once the reagent is mixed with the water sample, the colour development in the tubes is time-sensitive. Therefore, concentration ranges (according to colour development) should be recorded at exactly 3 minutes for nitrate and 5 minutes for phosphate.



The steps for collecting nitrate and phosphate measurements are as follows:

- Fill the measuring cup with the river water sample to the 1.5ml line.
- Remove the small yellow pin from the top of the testing tube to open the hole.
- Squeeze the sides of the tube to expel about half of the air volume.
- Whilst squeezing, insert the tube into the sample cup until it touches the bottom.
- Release the tube and allow the tube to suck up all of the sample water (the tube will be about half full).
- Place a finger over the pin hole and shake lightly until all of the powder is immersed in the water (its okay if the powder doesn't fully dissolve).
- Place the tube on a flat surface for the required time (nitrate = 3 minutes, phosphate = 5 minutes). Keep the tube in the palm of your hand on cold days (less than 10 °C).
- Place the tube between colours on the corresponding colour chart and observe the best match. Observations should be made out of direct sunlight.



This test should be repeated for both nitrate and phosphate.

The solution in the test tube should be disposed of away from the river, ideally the solution should be squeezed out in a sink and the plastic tube recycled. Please note that reactive powder is safely contained durable tubes, but should be stored away from non-trained individuals as it can be harmful (see the risk assessment for further details).

@brumriverchamps
birminghamriverchampions@gmail.com
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